CAI MILLS

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ABOUT ME:

I'm an enthusiastic learner who is always eager to advance my knowledge and skills, whether it's by refining my existing skills or broadening my understanding with innovative tools and programming languages. My favourite engine of choice for personal projects is Godot, specifically with C#.

SKILLS:

Languages:

- C++
- C#
- Python
- HTML & CSS
- Git

Engines / Frameworks:

- Godot
- Unity
- Unreal Engine
- SDL
- MonoGame
- DirectX 11
- OpenGL

Other Skills:

- Microsoft Office
- SQL (Brief)
- Collaboration
- Communication
- Problem Solving

HIGHLIGHTED PROJECTS:

I'm currently developing a rhythm game with a custom level editor in Godot C# for my final year project. Past projects include (see my portfolio website for more details):

Adrenaline Junkie (Unreal Engine 5) – Junior Programmer:

- Developed a complete framework for the gun system, including accuracy, bullet spread, reloading, etc.
- Implemented a conditional time-slow mechanic.
- Created several other miscellaneous features such as environmental traps, saving, loading, health regeneration and more.

Just Another Dialogue Editor (Godot) – Solo Project:

- Designed and built a node-based dialogue flowchart editor, allowing for the creation of serialisable conversations.
- Implemented JSON file serialisation and deserialisation.
- Created an accompanying dialogue system so that the serialised files can be tested within the project itself.

Deepest Dungeon (Unity) – Solo Project:

- Designed AI agents that can recognise the players' actions and respond with a few unique options.
- Created a weapon switching system, with each individual weapon having its own stats, such as damage, speed and range.
- Added stamina-based movement, including a dodge roll which moved the player forwards whilst providing temporary invulnerability.

EDUCATION:

University of Staffordshire
Computer Games Programming – BSc (Hons)
September 2023 to May 2026
Expected Grade: Upper Second – First Class

Notable Year 3 Modules:

- Low-Level Platform Optimisation (C++) – This involves utilising optimisation techniques such as memory management and spatial portioning to optimise a given scene, as well as porting a project using a PS5 Dev Kit.
- Advanced Game Artificial Intelligence (C++) – This involves learning different AI techniques such as goal-oriented action planning, genetic algorithms, adversarial search, and more.

WORK EXPERIENCE:

The Plough and Harrow (Swansea)
Waiter / Bar Staff
March 2023 to September 2023

Skills Learned:

- Communication
- Customer Service
- Teamwork
- Problem Solving
- How to manage in a work environment

HOBBIES:

Drawing | Table-top Role-Playing Games | Making Music

(References can be provided if requested)